LAB DOCUMENTATION

IPsec VPN

Hub and Spoke Topology

FortiGate Deployment on PNETLab

Item	Details
Project name	IPsec VPN – Hub and Spoke Topology
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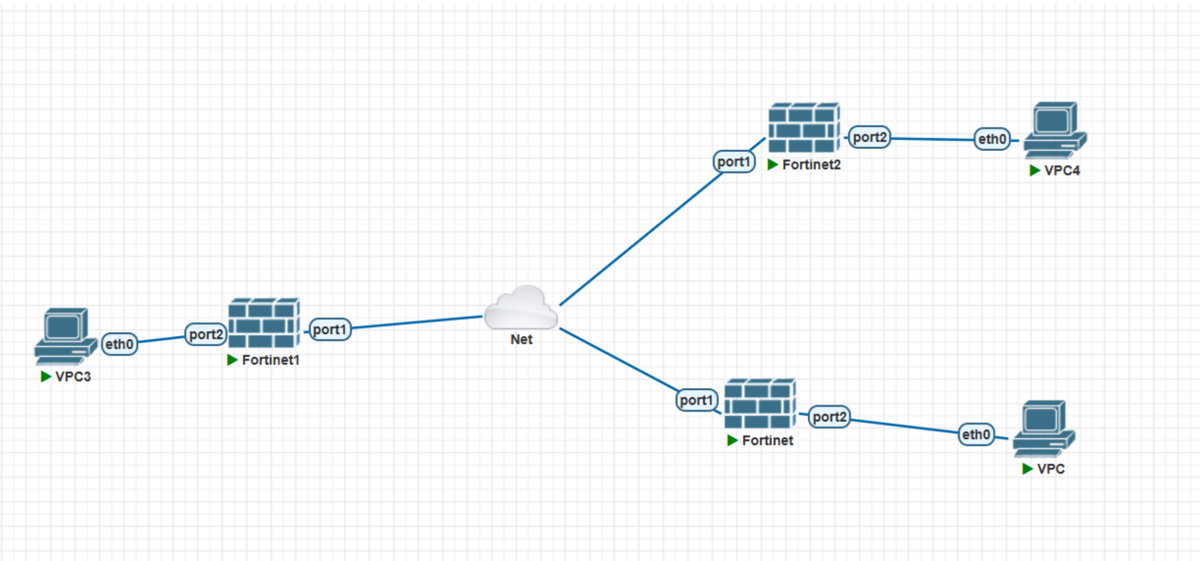
1. Project Overview

This document describes the setup and testing of an IPsec VPN connection using a Hub and Spoke topology, implemented with FortiGate virtual appliances inside the PNETLab simulation environment. The lab consists of one central FortiGate device acting as the Hub, and two FortiGate devices acting as Spokes (Spoke1 and Spoke2), so that traffic between the two branches (Spoke to Spoke) is routed through the central Hub device.

Network Topology



Device Role Mapping

Device Name	Role	Description
Fortinet	Hub	Central device connecting both branches
Fortinet1	Spoke 1	First branch connected to the Hub
Fortinet2	Spoke 2	Second branch connected to the Hub

2. Hub Configuration (Fortinet)

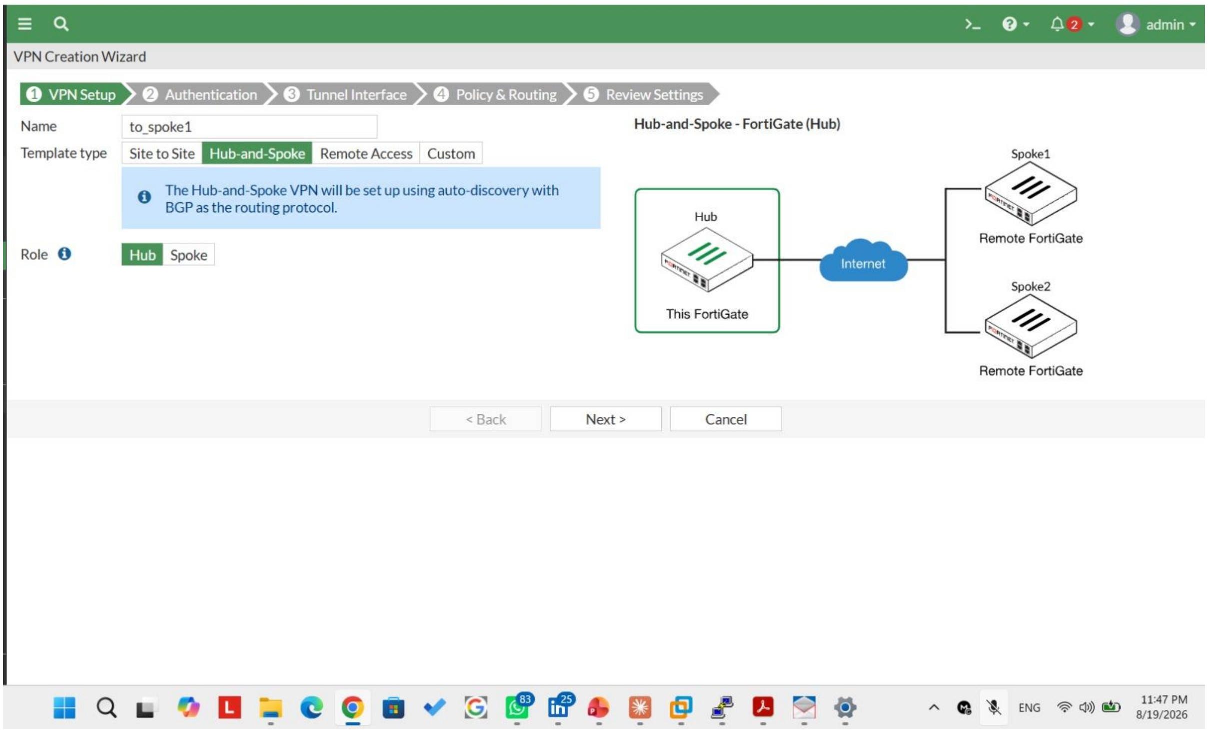
The Hub device was configured using the built-in Hub-and-Spoke wizard available in the FortiGate GUI, rather than the manual Custom configuration method. The steps and corresponding screens are documented below in order.

2.1 Accessing the VPN Setup Page

From the side menu: VPN → IPsec Tunnels → Create New, then select the Hub-and-Spoke template.

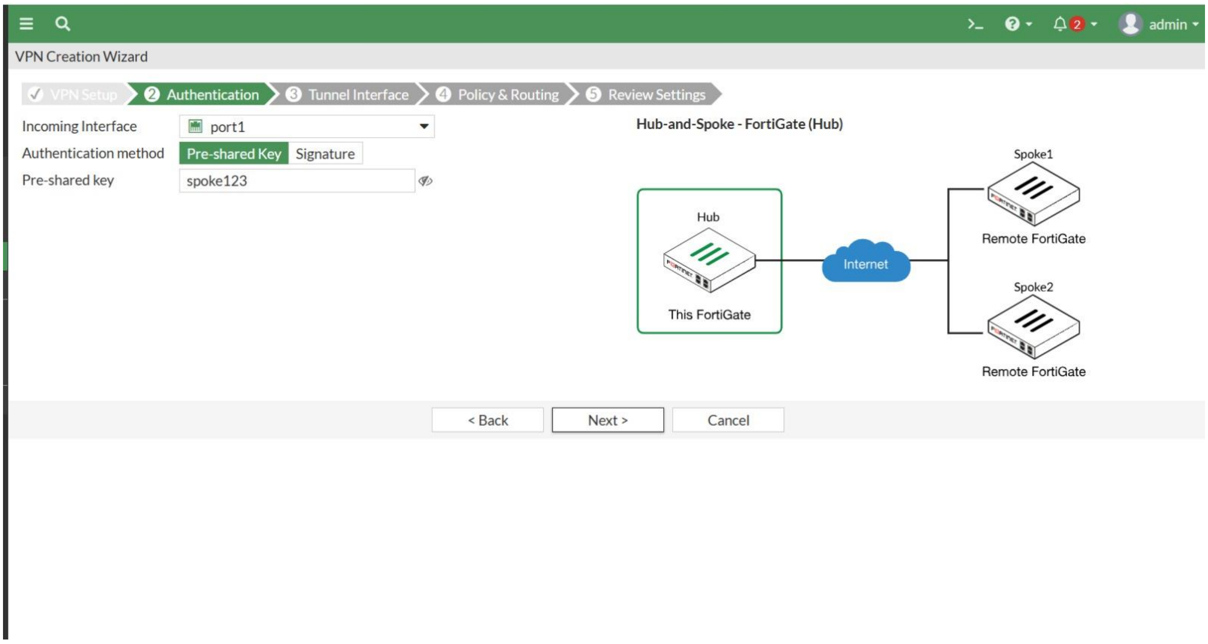
2.2 Basic Hub Configuration

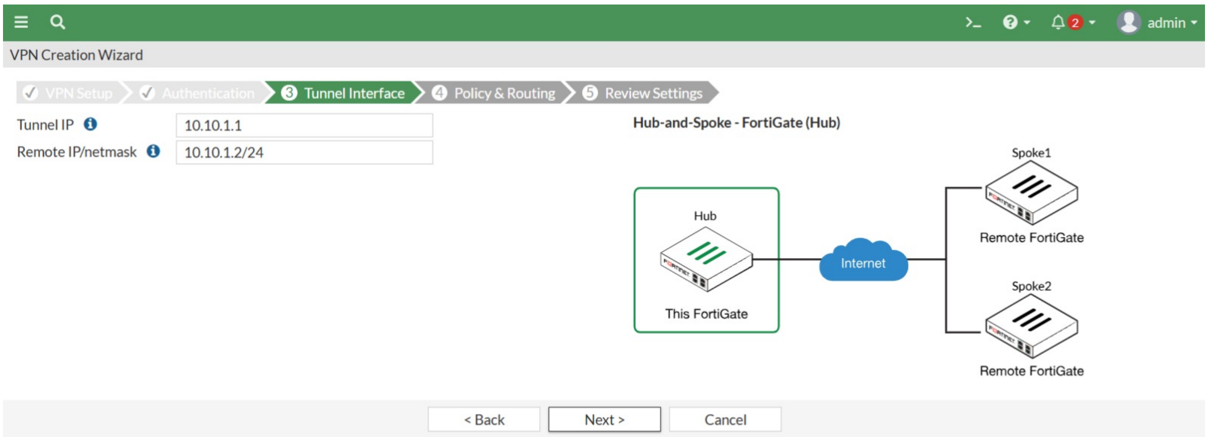
- Set the name of the Hub's tunnel.
- Select the outgoing (WAN) interface used to connect to the branches.
- Enter the Hub's internal (LAN) network.

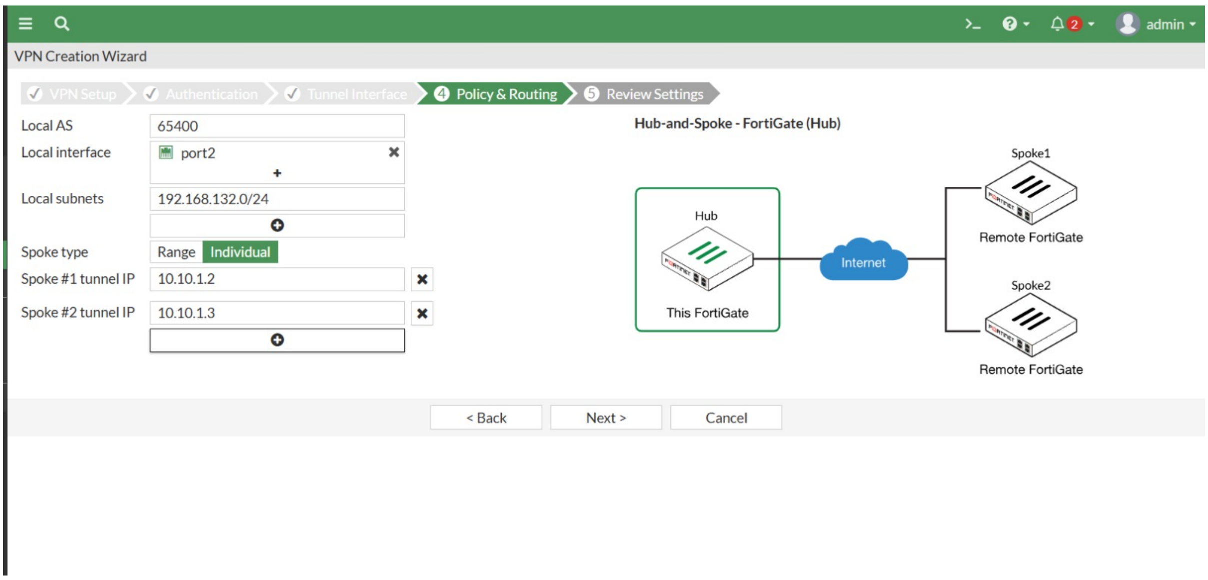


2.3 Authentication Settings

- Select Pre-shared Key as the authentication method.
- Select the IKE version to be used.

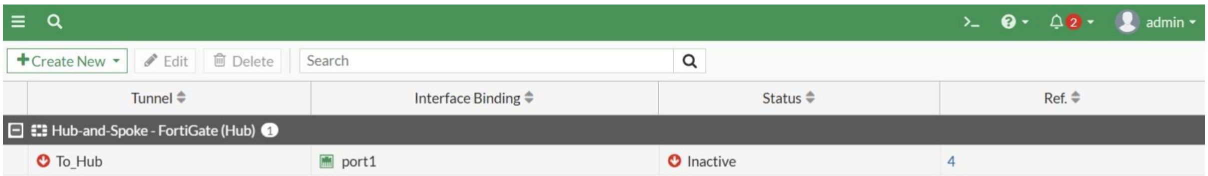

admin
VPN Creation Wizard
VPN Setup
2 Authentication
3 Tunnel Interface
4 Policy & Routing
5 Review Settings
Incoming Interface
port1
Authentication method
Pre-shared Key
Signature
Pre-shared key
spoke123
Hub-and-Spoke - FortiGate (Hub)
Hub
This FortiGate
Internet
Spoke1
Remote FortiGate
Spoke2
Remote FortiGate
< Back
Next >
Cancel


admin
VPN Creation Wizard
VPN Setup
Authentication
3 Tunnel Interface
4 Policy & Routing
5 Review Settings
Tunnel IP
10.10.1.1
Remote IP/netmask
10.10.1.2/24
Hub-and-Spoke - FortiGate (Hub)
Hub
This FortiGate
Internet
Spoke1
Remote FortiGate
Spoke2
Remote FortiGate
< Back
Next >
Cancel


admin
VPN Creation Wizard
VPN Setup
Authentication
Tunnel Interface
4 Policy & Routing
5 Review Settings
Local AS
65400
Local interface
port2
Local subnets
192.168.132.0/24
Spoke type
Range
Individual
Spoke #1 tunnel IP
10.10.1.2
Spoke #2 tunnel IP
10.10.1.3
Hub-and-Spoke - FortiGate (Hub)
Hub
This FortiGate
Internet
Spoke1
Remote FortiGate
Spoke2
Remote FortiGate
< Back
Next >
Cancel

3. Verifying the Tunnel on the Hub

After completing the Hub configuration, the VPN → IPsec Tunnels page was checked again to confirm that the Hub's tunnel was created successfully and appears in the list.



4. Spoke 1 Configuration (Fortinet1)

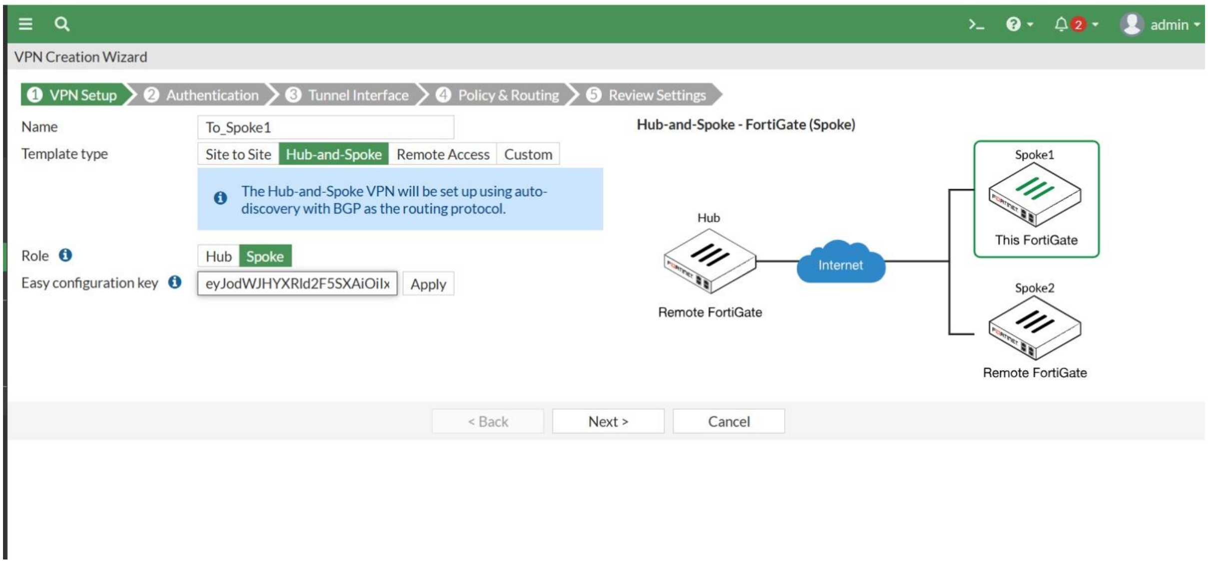
The same configuration steps used on the Hub were followed on Spoke1, selecting the device role as Spoke in the wizard instead of Hub, and entering the connection details specific to the first branch.

4.1 Accessing the VPN Setup Page on Spoke1

VPN → IPsec Tunnels → Create New → select the Spoke role template.

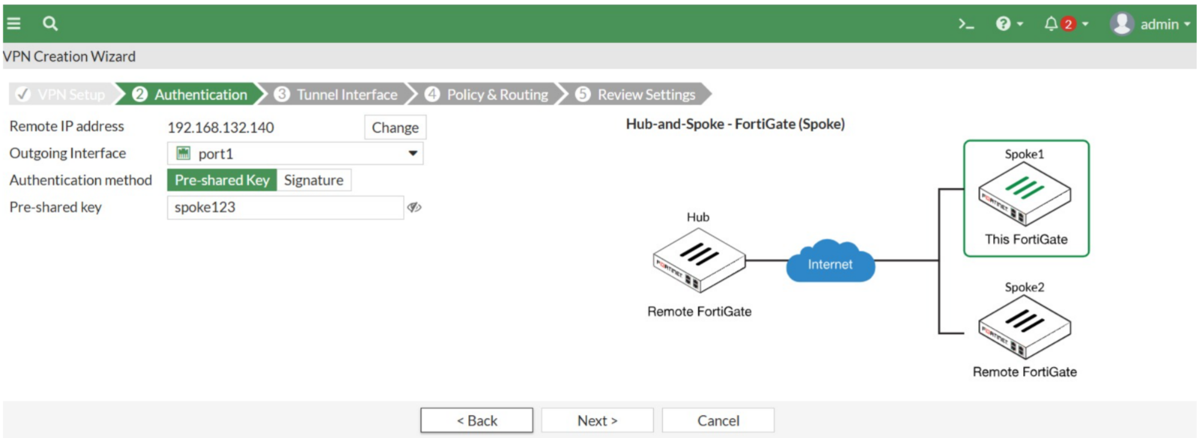
4.2 Connection Settings to the Hub

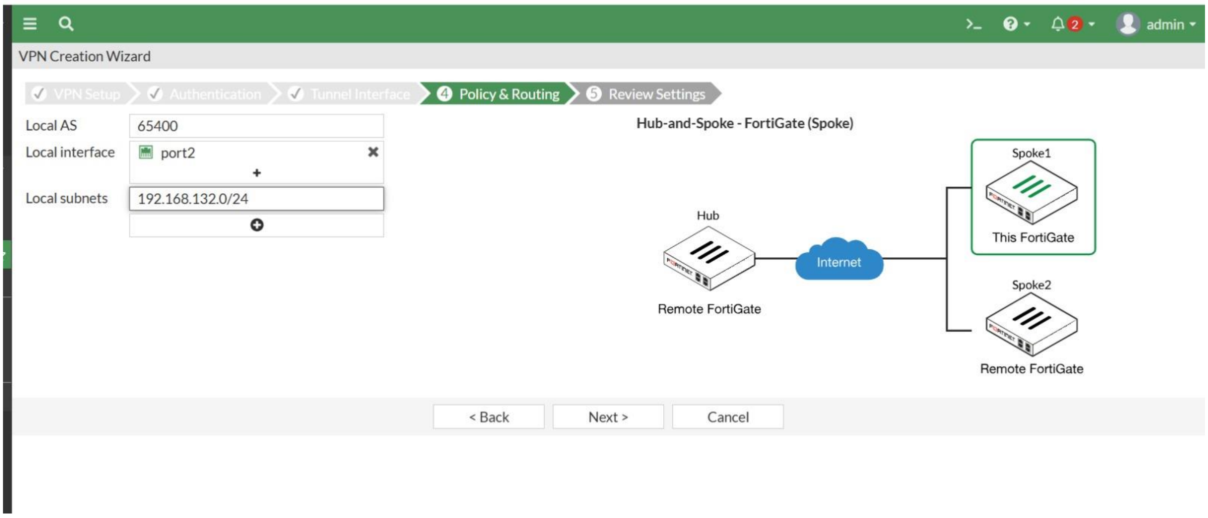
- Select the outgoing (WAN) interface on Spoke1.
- Enter the Hub's address as the Remote Gateway.
- Enter Spoke1's internal (LAN) network.



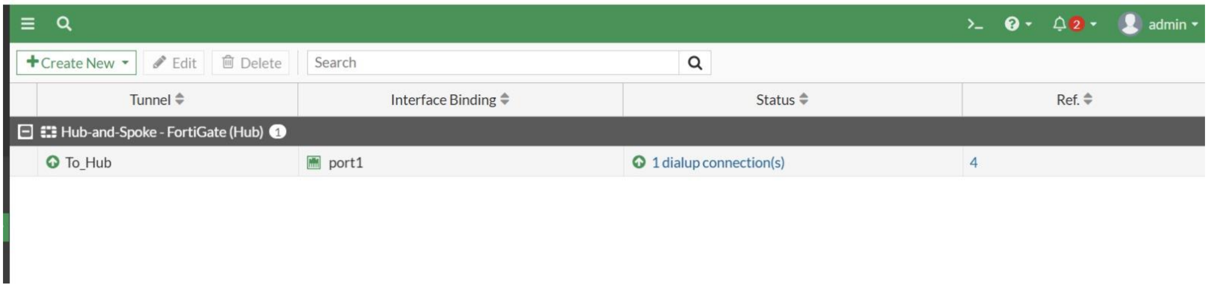
4.3 Authentication Settings on Spoke1

The same Pre-shared Key configured on the corresponding Hub side must be entered here to ensure the connection succeeds.





4.4 Reviewing Spoke1 Connection on Hub

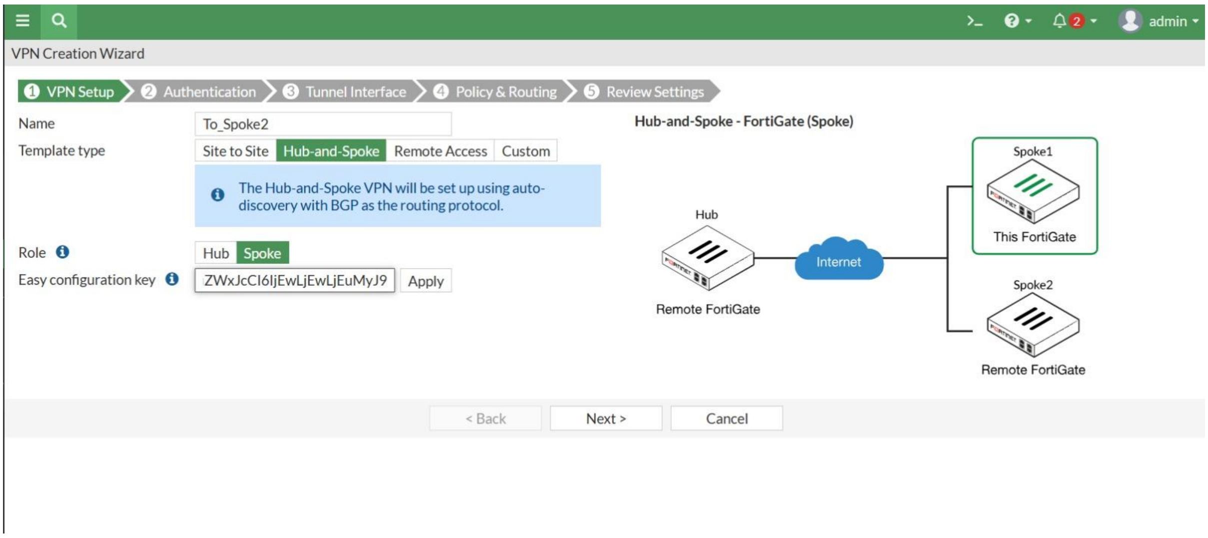


5. Spoke 2 Configuration (Fortinet2)

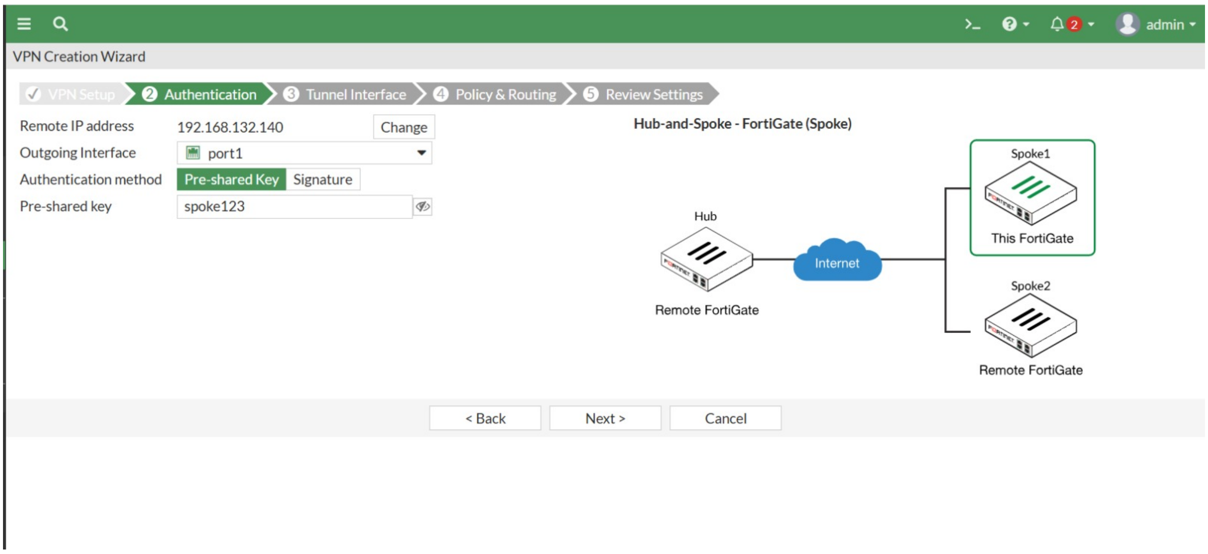
The same Spoke1 configuration steps were repeated in full on Fortinet2, updating the connection details to match the second branch.

5.1 Accessing the VPN Setup Page on Spoke2

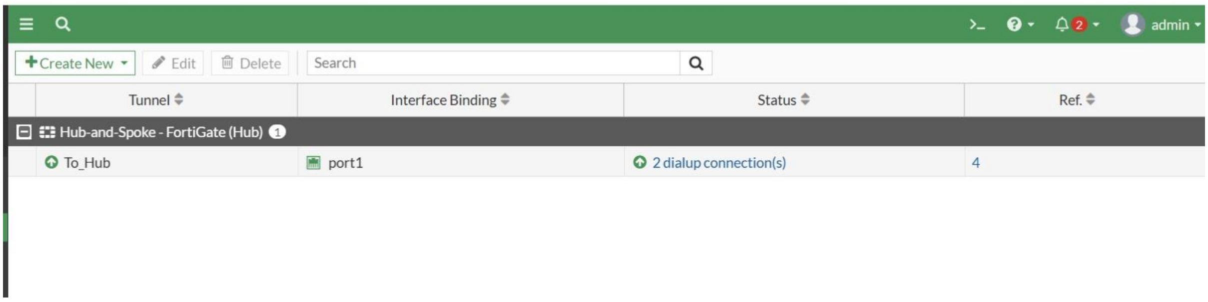
5.2 Connection Settings to the Hub



5.3 Authentication Settings on Spoke2



5.4 Reviewing Spoke2 Connection on Hub



6. Verifying Tunnel Status via CLI

To confirm that the tunnels were actually Up and functioning correctly, the following command was run through the CLI console embedded in the GUI (not via external SSH):

```
diagnose vpn tunnel list
```

6.1 Command Output on Spoke1

```
CLI Console (1)
BR-1 # diagnose vpn tunnel list
list all ipsec tunnel in vd 0
------------------------------------------------------
name=To_Spoke1 ver=1 serial=1 192.168.132.139:0->192.168.132.140:0 tun_id=192.168.132.140 tun_id6=::192.168.132.140 dst_mtu=1500 dpd-lin
k=on weight=1
bound_if=3 lgwy=static/1 tun=tunnel/255 mode=auto/1 encap=none/528 options[0210]=create_dev frag-rfc  accept_traffic=1 overlay_id=0
proxyid_num=1 child_num=0 refcnt=4 ilast=19 olast=19 ad=r/2
stat: rxp=35 txp=24 rxb=4072 txb=1536
dpd: mode=on-idle on=1 idle=20000ms retry=3 count=0 seqno=10
natt: mode=none draft=0 interval=0 remote_port=0
proxyid=To_Spoke1 proto=0 sa=1 ref=3 serial=1 adr
  src: 0:0.0.0.0/0.0.0.0:0
  dst: 0:0.0.0.0/0.0.0.0:0
  SA:  ref=3 options=32202 type=00 soft=0 mtu=1446 expire=42463/0B replaywin=2048
       seqno=19 esn=0 replaywin_lastseq=00000024 itn=0 qat=0 hash_search_len=1
  life: type=01 bytes=0/0 timeout=42902/43200
  dec: spi=980f02b0 esp=des key=8 9de922d7cda13774
       ah=md5 key=16 54dd04396fbe35ee5ff30b7bbc4d31af
  enc: spi=e1946681 esp=des key=8 1ec46525d7992be0
       ah=md5 key=16 9917e782d7395049945f43b5ae443421
  dec:pkts/bytes=35/2204, enc:pkts/bytes=24/2832

BR-1 #
```

6.2 Command Output on Spoke2

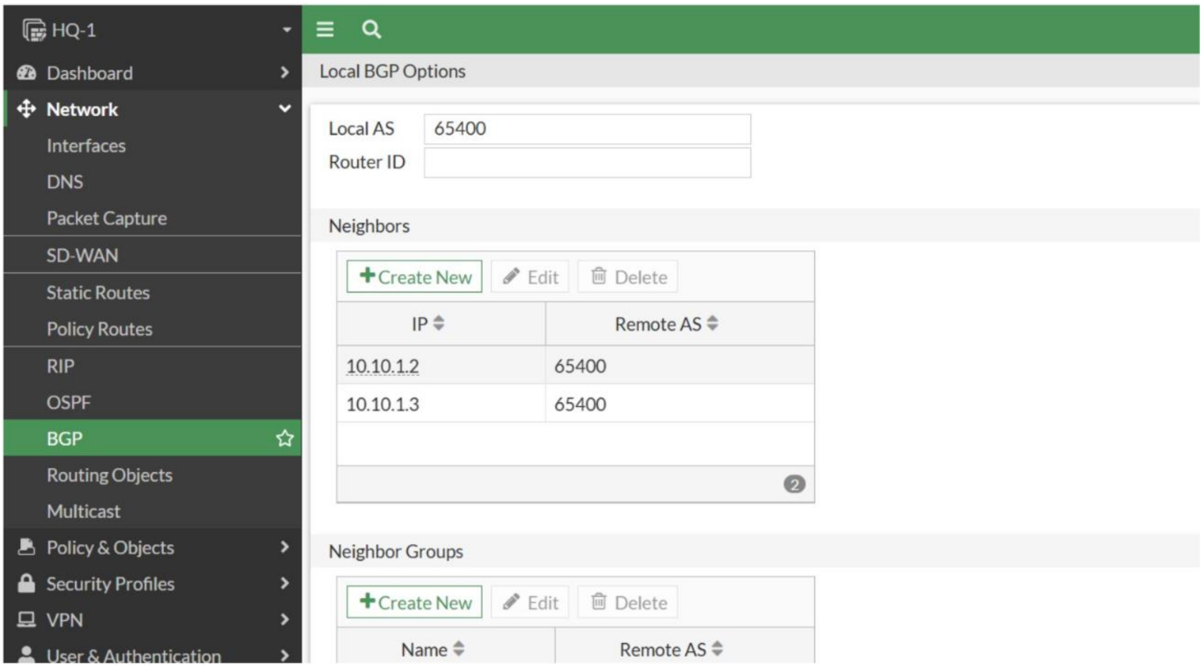
```
CLI Console (1)
BR-2 # diagnose vpn tunnel list
list all ipsec tunnel in vd 0
------------------------------------------------------
name=To_Spoke2 ver=1 serial=1 192.168.132.141:0->192.168.132.140:0 tun_id=192.168.132.140 tun_id6=::192.168.132.140 dst_mtu=1500 dpd-lin
k=on weight=1
bound_if=3 lgwy=static/1 tun=tunnel/255 mode=auto/1 encap=none/528 options[0210]=create_dev frag-rfc  accept_traffic=1 overlay_id=0
proxyid_num=1 child_num=0 refcnt=4 ilast=10 olast=10 ad=r/2
stat: rxp=11 txp=12 rxb=1336 txb=798
dpd: mode=on-idle on=1 idle=20000ms retry=3 count=0 seqno=2
natt: mode=none draft=0 interval=0 remote_port=0
proxyid=To_Spoke2 proto=0 sa=1 ref=3 serial=1 adr
  src: 0:0.0.0.0/0.0.0.0:0
  dst: 0:0.0.0.0/0.0.0.0:0
  SA:  ref=3 options=32202 type=00 soft=0 mtu=1446 expire=42767/0B replaywin=2048
       seqno=d esn=0 replaywin_lastseq=0000000c itn=0 qat=0 hash_search_len=1
  life: type=01 bytes=0/0 timeout=42903/43200
  dec: spi=df60d7a1 esp=des key=8 c7703c310d482fa5
       ah=md5 key=16 50e876220e5c655d63793bcd211bbd78
  enc: spi=e1946682 esp=des key=8 8ed7923fb0138f96
       ah=md5 key=16 190e1646e5e28d08637566b53ccee37f
  dec:pkts/bytes=11/746, enc:pkts/bytes=12/1440

BR-2 #
```

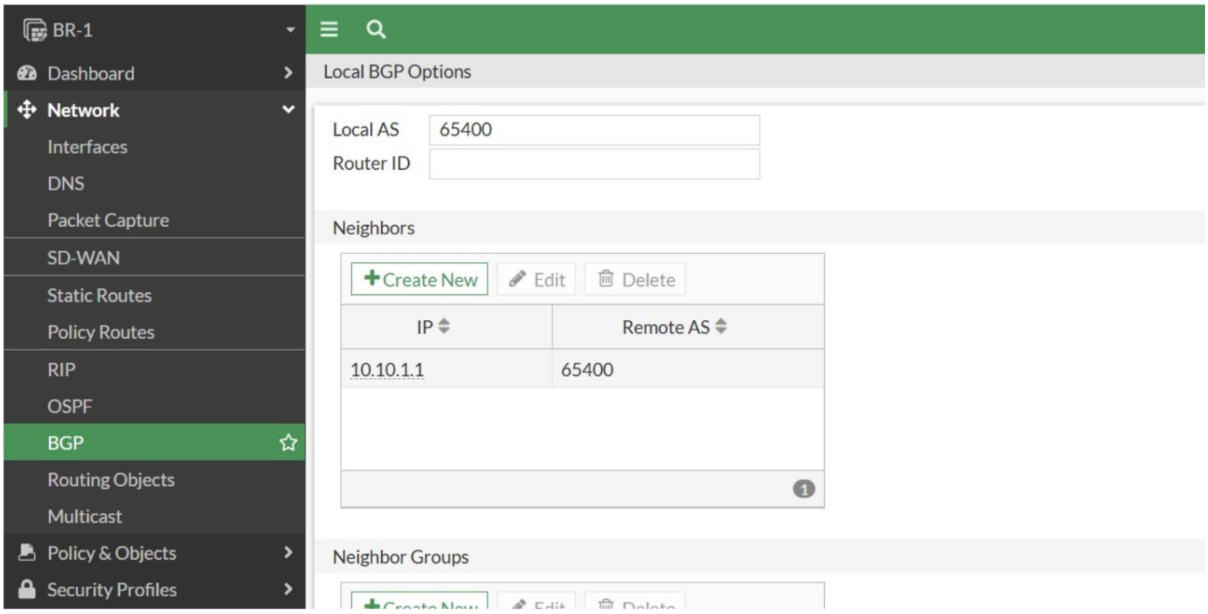
7. Monitoring Connection Status via the Network Page

As a final verification step, the Network section was accessed from the side menu, followed by the BGP page, to monitor the session status on each of the three devices.

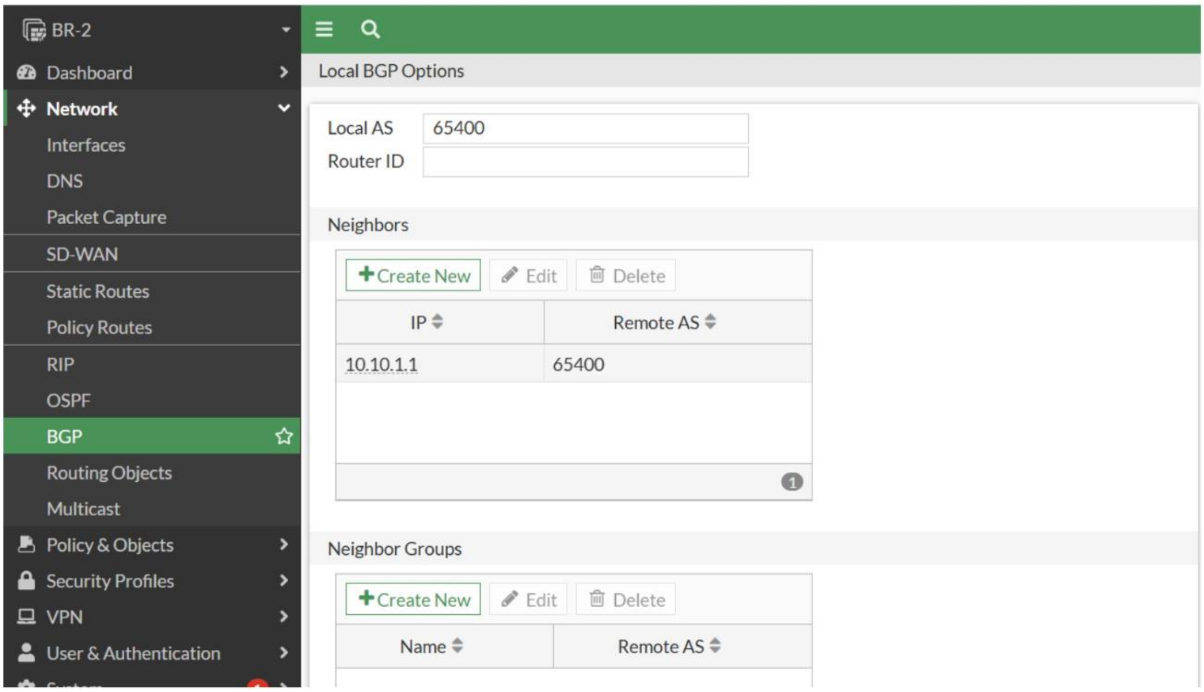
7.1 BGP Status on the Hub



7.2 BGP Status on Spoke1



7.3 BGP Status on Spoke2



8. Conclusion

Through the steps above, an IPsec VPN connection was successfully built using a Hub and Spoke topology between three FortiGate devices inside the PNETLab environment, with traffic between the two branches (Spoke1 and Spoke2) routed through the central Hub device. The success of the connection was verified through the GUI tunnel status, the 'diagnose vpn tunnel list' CLI command, and by monitoring the session status on the Network page.